

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

__show ip route ospf__

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _Serial0/0/0_

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _129_

Does R2 show up as an OSPF neighbor on R1? _yes_

Does R2 show up as an OSPF neighbor on R3? _no_

What does this information tell you?

R2 only has OSPF adjacency with R1, but not with R3, since R3 learns about the 192.168.2.0/24 network through R1 via OSPF, using a path through R1 rather than a direct connection to R2.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

router ospf 1

network 192.15=68.23.0 0.0.0.3 area 0

end

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _Serial0/0/1_

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65, R3 to R2 cost = 64 -> R2 to G0/0 cost = 1 and Total = 64 + 1 = 65

Is R2 listed as an OSPF neighbor to R3? _Yes_

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To allow OSPF to distinguish between Fast Ethernet, Gigabit Ethernet, and higher-speed links so it can make more accurate routing decisions based on actual interface bandwidth.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24 is 781 (R1 S0/0/1) + 1 (R3 G0/0) = 782.

The cost to 192.168.23.0/30 is 781 (R1 serial) + 64 (R3 S0/0/1) = 845.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

Because all serial links now have bandwidth 128 Kbit, which each serial interface an OSPF cost of 781. The total cost is 781 (R1) + 781 (R3) = 1562.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The "ip ospf cost 1565" command on R1's S0/0/1 made the direct path to 192.168.3.0/24 cost 1566, which is higher than the path through R2 costing 1563. OSPF automatically selects the lowest-cost path, so traffic now goes through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Controlling router ID can ensure consistent router identification, prevent OSPF neighbor issues and avoid unexpected router ID changes that can disrupt adjacencies.

2. Why is the DR/BDR election process not a concern in this lab?

No DR/BDR election because all connections are point-to-point serial links, so the network type uses P2P instead of broadcast.

3. Why would you want to set an OSPF interface to passive?

Setting an interface to passive can prevent OSPF from sending hello messages there, stop unnecessary adjacencies and advertise the connected network.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF uses areas to divide networks into smaller sections, limiting route updates and reducing routing table size.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

"network 10.0.0.0 0.255.255.255 area 0"

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

~~The OSPF route~~ will be used because OSPF has the lowest administrative distance (110) compared to RIP (120) and EIGRP (170).